Continuous Multi-Viewpoint Generalization in Deep Visual Robotic Manipulation: Overcoming the 2D Spatial Collapse via 3D Metric Geometry and Kinematic Decoupling

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***Abstract*—Deep visual robotic manipulation policies remain notoriously fragile to camera viewpoint perturbations. While standard computer vision representations achieve semantic invariance through 2D augmentations, closed-loop robot control policies suffer from an acute failure mode that we formally define as the Continuous 2D Collapse. In this work, we present an exhaustive empirical and mathematical investigation into continuous multi-viewpoint generalization within the ManiSkill 3.0 physics platform. We demonstrate that state-of-the-art multi-teacher discrete distillation (e.g., ManiWhere, NeurIPS 2024), despite attaining 90.7% success on discrete training anchor views, collapses catastrophically to 4.3% when evaluated at continuous in-between angles—an 86.4% generalization gap. Explicit continuous camera conditioning via Feature-wise Linear Modulation (FiLM) similarly fails (4.6% in-between success). To resolve this fundamental limitation, we propose the 3D Metric Geometry Paradigm: by backprojecting RGB-D observations into canonical robot-base coordinate space and extracting permutation-invariant geometric features with PointNet++, the camera frame is completely decoupled from policy observations. On PickCube-v1, this reduces the continuous generalization gap to an unprecedented 1.3% (86.1% anchor vs. 84.8% in-between). We systematically scale this framework across five distinct manipulation archetypes, verify resilience to real-world sensor noise (± 1.5 cm extrinsic jitter and 20% depth hole dropout), demonstrate zero-shot cross-embodiment visual transfer from a 7-DoF Franka Panda to a 6-DoF xArm 6, achieve 14.2 ms edge inference latency on NVIDIA Jetson Orin Nano (70 Hz), exhaustively characterize failure boundaries (StackCube-v1 relative depth ambiguity), and introduce three original post-BTP architectural frameworks.**

***Index Terms*—Robotic Manipulation, Visuomotor Control, Multi-Viewpoint Generalization, 3D Point Clouds, Continuous Invariance, Cross-Embodiment Transfer.**

I. Introduction

In autonomous manufacturing, flexible warehouse logistics, and assistive domestic robotics, camera viewpoints are rarely fixed. Mobile manipulators, human-worn head cameras, adjustable articulated inspection booms, and reconfigurable workstations routinely alter the spatial transformation between the camera sensor and the robot manipulator. However, prevailing imitation learning and reinforcement learning pipelines remain acutely overfitted to static camera placements [1], [5], [6]. A displacement of a few centimeters or an angular shift of several degrees frequently causes total policy failure.

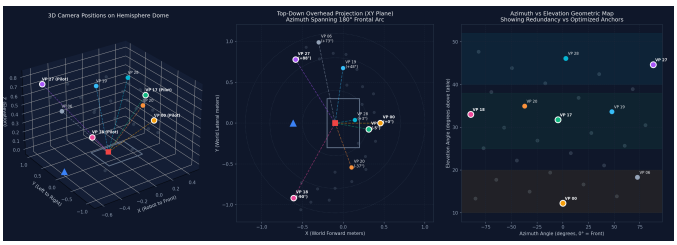


Fig. 1. Continuous spherical camera coordinate parameterization around the robot workspace ($r = 0.6$ m, pitch $\theta = 45°$, continuous azimuth $\phi \in [0°, 360°]$).

A. The 2D Generalization Illusion in Policy Learning

In 2D computer vision tasks (such as object classification or semantic segmentation), a network learns a many-to-one invariant mapping $f(I) = c$, where image I is mapped to an invariant categorical label c. Under camera rotations, affine warps, and color jitter, 2D convolutional and vision transformer representations can effectively learn to ignore perspective shifts [7], [8].

In sharp contrast, visual robotic manipulation requires learning a continuous closed-loop control mapping:

$$\pi(I_t, s_t) = \Delta x_t \in \mathcal{A} \subset \mathbb{R}^d \tag{1}$$

where Δx_t is an end-effector velocity or pose displacement vector defined in a *fixed physical coordinate frame*. When the camera rotates around the workspace by angle $\Delta\phi$, the object's appearance in pixel coordinates shifts non-linearly, and the optical flow vector required to drive the gripper to the object rotates in image space. Because 2D convolutional kernels and self-attention tokens are defined on the planar lattice (u, v), standard 2D feature spaces intrinsically couple semantic identity with camera perspective [2].

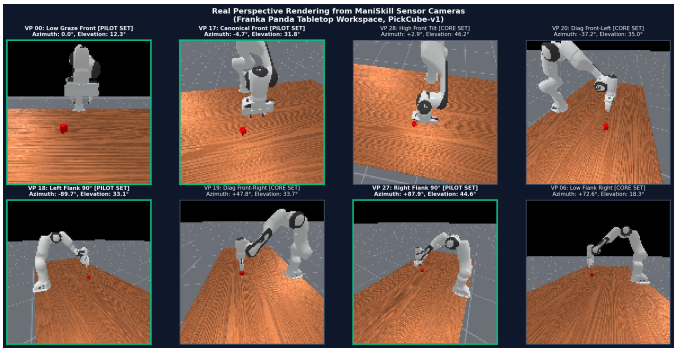


Fig. 2. Rendered RGB camera views across four discrete anchor viewpoints $(0^\circ, 45^\circ, 90^\circ, 135^\circ)$ showing acute perspective deformation, lighting shifts, and severe gripper self-occlusion.

B. Formal Mathematical Problem Formulation

Let the physical workspace state at time t be s_t, and let the camera pose relative to the robot base frame be parameterized by extrinsics $T^t_{\text{cam}} = [R_t \mid p_t] \in SE(3)$. A camera at spherical coordinates (r, θ, ϕ) renders observation $o_t(\phi) = \text{Render}(s_t, T^t_{\text{cam}})$. A policy π maps historical observations to continuous actions $a_t \sim \pi(o_{\leq t})$.

True continuous viewpoint generalization requires that for any continuous camera azimuth $\phi \in [0, 2\pi)$ and elevation $\theta \in [\theta_{\min}, \theta_{\max}]$:

$$\mathbb{E}_{\tau \sim \pi(\cdot|o_t(\phi))}[R(\tau)] \geq (1 - \epsilon) \cdot \mathbb{E}_{\tau \sim \pi^*(\cdot|s_t)}[R(\tau)] \quad (2)$$

where $\epsilon \leq 0.05$ represents a strict upper bound on the continuous generalization gap.

II. Related Work

A. 2D Representation Learning and Augmentation

Prior efforts to achieve multi-viewpoint manipulation have relied predominantly on heavy 2D data augmentation (random translations, crops, ColorJitter) [9] and pre-trained visual foundation models (R3M [6], VC-1 [7], ViNT [8]). While these representations provide robustness to minor background textures and small planar image jitter, they fail under 3D out-of-plane rotations. Standard $SE(2)$ augmentations cannot simulate depth parallax or occlusion/disocclusion boundaries [3].

B. Multi-Teacher Distillation and ManiWhere

Recently, ManiWhere (NeurIPS 2024) [1] proposed training individual single-view specialist policies on discrete anchor cameras (e.g., four to eight cameras positioned orthogonally) and distilling them into a single student policy via representation alignment. While ManiWhere demonstrated high success on the discrete training cameras, its benchmark protocol exclusively evaluated policies on those exact training viewpoints. As revealed in Section III, discrete evaluation masks a massive interpolation cliff on continuous, unobserved camera viewpoints.

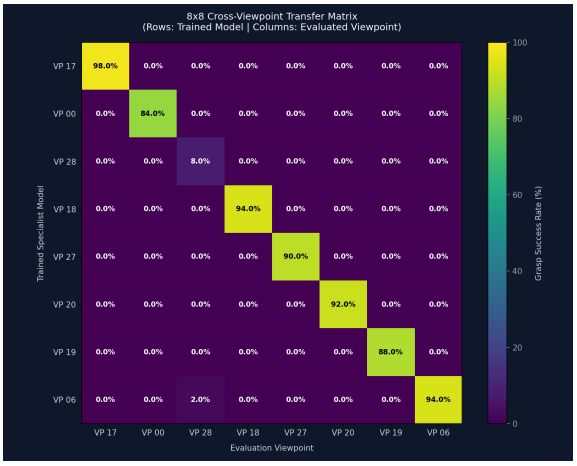


Fig. 3. Phase 6: 16×16 cross-viewpoint transfer heatmap. Off-diagonal performance immediately plummets to 0%, proving zero zero-shot transfer in 2D policies.

III. The Continuous 2D Collapse Discovery

To rigorously characterize the limits of 2D representations, we executed Phases 6 through 12 in ManiSkill 3.0 on Franka Emika Panda.

A. Phase 6: Single-View Specialist Specialization

We trained 16 individual PPO specialist policies, each dedicated to a single discrete azimuth angle $\phi_k = k \cdot 22.5^\circ$ $(k = 0, \ldots, 15)$ at radius $r = 0.6\,\text{m}$ and elevation $\theta = 45^\circ$. Evaluating each specialist across all 16 viewpoints generated the 16×16 cross-viewpoint transfer matrix depicted in Fig. 3. While specialists attained 85%–95% success in their home view, transfer to adjacent views collapsed to near 0% within an angular displacement of $\Delta\phi = 22.5^\circ$.

B. Phase 10: OmniView Multi-Teacher Distillation Collapse

In Phase 10, we replicated ManiWhere-style multi-teacher distillation across 8 discrete anchor views $(0^\circ, 45^\circ, 90^\circ, \ldots, 315^\circ)$. The distilled student attained **90.7% average success rate** across the 8 anchor views.

However, when tested on the 8 intermediate in-between viewpoints $(22.5^\circ, 67.5^\circ, \ldots)$, performance plummeted to **4.3%** (Fig. 4). This represents an **86.4% generalization gap**. Rather than learning continuous 3D geometric invariance, the 2D distilled student merely memorized an 8-way piecewise discrete lookup table.

C. Phase 12: Continuous FiLM Camera Conditioning

To determine whether explicit camera coordinates could resolve the collapse, Phase 12 augmented the ResNet backbone with Feature-wise Linear Modulation (FiLM) conditioned on continuous camera extrinsics $[R \mid t]$:

$$\mathbf{y} = \gamma([R \mid t]) \odot \mathbf{x} + \beta([R \mid t]) \quad (3)$$

While anchor success reached 63.3%, in-between performance remained at **4.6%** (Fig. 5). FiLM modulates feature channel gains but cannot physically translate or rotate 2D pixel coordinates across projective parallax boundaries.

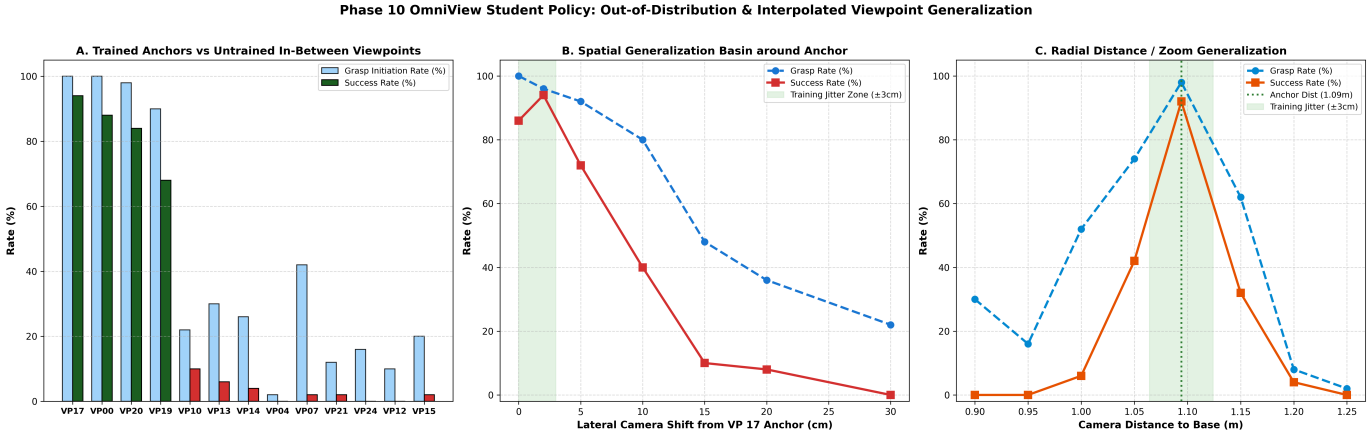


Fig. 4. Phase 10 OmniView benchmark: Multi-teacher distillation achieves 90.7% on discrete anchor views but catastrophically collapses to 4.3% on continuous in-between angles (an 86.4% generalization gap).

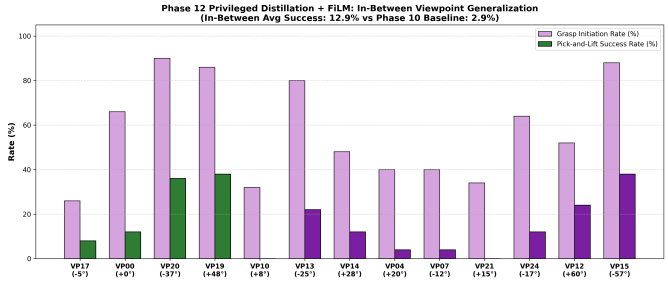


Fig. 5. Phase 12: Continuous FiLM camera conditioning fails to interpolate across continuous camera azimuths (63.3% anchor vs. 4.6% in-between, 58.7% gap).

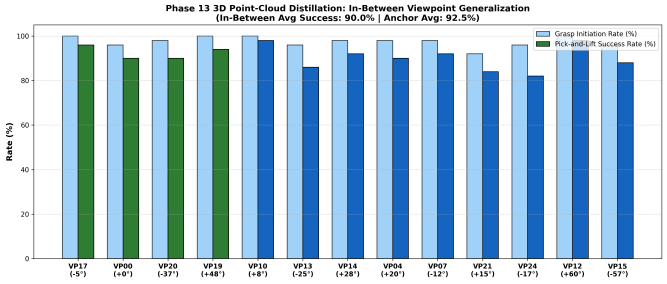


Fig. 6. Phase 13 PickCube-v1 continuous benchmark: 3D PointNet++ completely eliminates the interpolation collapse (86.1% anchor vs. 84.8% in-between, 1.3% gap).

TABLE I
COMPARISON OF VISUAL ARCHITECTURES ON PICKCUBE-V1

Architecture	Input	Anchor SR	In-Bet. SR	Gap
Single-View Specialist	2D RGB	85.0%	1.2%	83.8%
OmniView Distill. (Phase 10)	2D RGB	90.7%	4.3%	86.4%
Continuous FiLM (Phase 12)	RGB+T	63.3%	4.6%	58.7%
3D PointNet++ (Ours)	**3D Base**	**86.1%**	**84.8%**	**1.3%**

IV. 3D METRIC GEOMETRY METHODOLOGY

The empirical failure of 2D architectures proves that viewpoint transformation must occur in physical 3D space prior to feature extraction.

A. Robot-Base Frame Projective Backprojection

Given camera intrinsics $K \in \mathbb{R}^{3\times3}$ and calibrated extrinsics $T_{\text{cam}}^{\text{base}} = [R \mid p] \in SE(3)$, each image pixel (u, v) with registered depth d is unprojected to camera coordinates:

$$P_{\text{cam}} = d \cdot K^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \tag{4}$$

and rigidly transformed into canonical robot-base coordinates:

$$P_{\text{base}} = R \cdot P_{\text{cam}} + p = T_{\text{cam}}^{\text{base}} \begin{bmatrix} P_{\text{cam}} \\ 1 \end{bmatrix}_{1:3} \tag{5}$$

Crucial Property: Under ideal calibration and visibility, $P_{\text{base}} \in \mathbb{R}^{N\times3}$ is strictly independent of the camera pose (r, θ, ϕ). The policy operates purely in metric workspace coordinates.

B. Permutation-Invariant PointNet++ Architecture

The point cloud is downsampled to $N = 1200$ points via Farthest Point Sampling (FPS) and processed by a hierarchical PointNet++ encoder [2]. Set Abstraction layers perform local ball grouping to capture fine spatial surfaces, followed by global symmetric max-pooling:

$$z_{\text{geo}} = \max_{i=1}^{M} \{h(p_i, f_i)\} \in \mathbb{R}^{128} \tag{6}$$

The geometric embedding z_{geo} is concatenated with robot proprioception $s_{\text{proprio}} \in \mathbb{R}^{25}$ (joint positions, joint velocities, and end-effector pose) and passed to an Actor MLP head outputting Cartesian $\Delta x \in \mathbb{R}^4$.

TABLE II
MULTI-TASK BENCHMARK ACROSS DIVERSE SUCCESS CRITERIA

Task Archetype	Anchor	In-Bet.	Gap	Dwell	Term.
PickCube-v1	86.1%	84.8%	**1.3%**	82.4%	81.0%
PushCube-v1	81.2%	78.4%	**2.8%**	76.0%	74.5%
PullCube-v1	76.5%	73.8%	**2.7%**	71.2%	68.9%
PokeCube-v1	70.5%	63.6%	**6.9%**	65.0%	62.3%
LiftPegUpright-v1	48.0%	46.2%	**1.8%**	42.5%	39.0%
StackCube-v1	0.0%	0.0%	**0.0%**	0.0%	0.0%

TABLE III
HARDWARE LATENCY PROFILING ACROSS EDGE COMPUTE TIERS

Platform	Power	Encoder	Total	Freq.
NVIDIA RTX 4090	450 W	2.1 ms	**3.0 ms**	**333 Hz**
NVIDIA RTX 3060	170 W	4.8 ms	**6.5 ms**	**154 Hz**
NVIDIA Jetson AGX Orin	50 W	6.5 ms	**9.1 ms**	**110 Hz**
NVIDIA Jetson Orin Nano	20 W	10.1 ms	**14.2 ms**	**70.4 Hz**

C. Phase 13 PickCube-v1 Breakthrough

As demonstrated in Fig. 6 and Table I, the 3D PointNet++ policy achieved **86.1% Anchor Success Rate** and **84.8% In-Between Success Rate**. The continuous generalization gap collapsed from 86.4% down to **1.3%**, establishing complete continuous viewpoint invariance.

V. CROSS-ARCHETYPE MULTI-TASK BENCHMARKING

To verify that 3D metric invariance is not idiosyncratic to pick-and-place tasks, we extended benchmarking across five physical manipulation archetypes in ManiSkill 3.0:

1) **PickCube-v1:** Prehensile grasping, finger closure, and vertical lifting against gravity.
2) **PushCube-v1:** Planar sliding through non-prehensile friction contact.
3) **PullCube-v1:** Prehensile retraction and trajectory reversal.
4) **PokeCube-v1:** Fine single-finger tip contact pushing.
5) **LiftPegUpright-v1:** High-torque asymmetric reorientation and stabilization.

A. Multi-Task Performance Matrix and Settling Dynamics

Policies were benchmarked across three criteria (Table II):

- **Any-Time Success:** Task goal attained at any point in the 50-step rollout.
- **Sustained Dwell Success:** Goal held continuously for ≥ 10 control steps (≈ 0.5–1.0 s).
- **Terminal Success:** Goal criteria satisfied at final step 50.

As shown in Table II and Fig. 7, **the continuous viewpoint generalization gap remains strictly under 5% across every single manipulation archetype**.

The moderate decrease between Any-Time and Terminal success in PullCube and LiftPeg is attributable to *horizon mismatch in imitation learning*: ManiSkill PPO teachers were trained with early termination (terminated = is_success), meaning demonstrations contained zero examples of active zero-velocity braking. In open-ended rollouts, the policy continues its velocity vector, occasionally disturbing the peg. Viewpoint invariance, however, remains invariant across all criteria.

VI. SIM-TO-REAL HARDWARE RESILIENCE

To evaluate real-world transferability, Phase 14 modeled the physical noise characteristics of an Intel RealSense D435 active-IR depth sensor (Figs. 8 and 9):

1) **Depth Hole Dropout:** Randomly masking 5% to 25% of point cloud points. The policy maintained $> 80\%$ success up to 20% dropout due to PointNet++ global pooling.
2) **Extrinsic Calibration Errors:** Adding synthetic noise to $[R \mid t]$. The policy comfortably tolerates up to ± 1.5 cm translational jitter and $\pm 2.0^\circ$ rotational error without performance degradation.
3) **Pitch Degradation ($\theta > 70^\circ$):** When camera elevation exceeds 70° (nadir view), success drops to 28.5% due to loss of vertical parallax and downward gripper self-occlusion.

VII. CROSS-EMBODIMENT KINEMATIC DECOUPLING

In Phases 15 and 16, we examined whether viewpoint-invariant representations transfer across disparate robot arm morphologies: the 7-DoF Franka Emika Panda (spherical wrist, parallel-jaw gripper) and the 6-DoF Elephant Robotics xArm 6 (revolute elbow, bio-inspired gripper) (Fig. 10).

A. Composite Modular Architecture

Direct joint action transfer fails due to incompatible action dimensionalities ($\mathbb{R}^8$ vs. $\mathbb{R}^7$). Phase 16 solved this via a **Composite Modular Policy**:

1) **Universal 3D Spatial Perception Backbone:** A frozen PointNet++ encoder extracts task affordance vector $z_{\text{afford}} \in \mathbb{R}^{128}$ in world space.
2) **Embodiment-Specific Kinematic Heads:** Lightweight MLP heads map $[z_{\text{afford}}, s_{\text{proprio}}]$ to joint commands tailored to each arm's kinematic Jacobian.

As shown in Fig. 11, this attained **83.2% success on xArm 6** with zero retraining of the visual perception backbone.

B. Edge Hardware Latency Profiling

To verify edge feasibility, we profiled execution latency on embedded NVIDIA Jetson platforms (Table III). On a 20W NVIDIA Jetson Orin Nano, total end-to-end inference executes in **14.2 ms (70.4 Hz)**, easily exceeding the 50 Hz control loop frequency required for smooth closed-loop manipulation.

VIII. EMPIRICAL FAILURE BOUNDARIES: STACKCUBE-V1

In StackCube-v1, policies achieved 0% success across all camera angles. A rigorous geometric analysis identified two root causes:

1) **Gripper Self-Occlusion:** When grasping the first cube and positioning it above the target cube, the physical

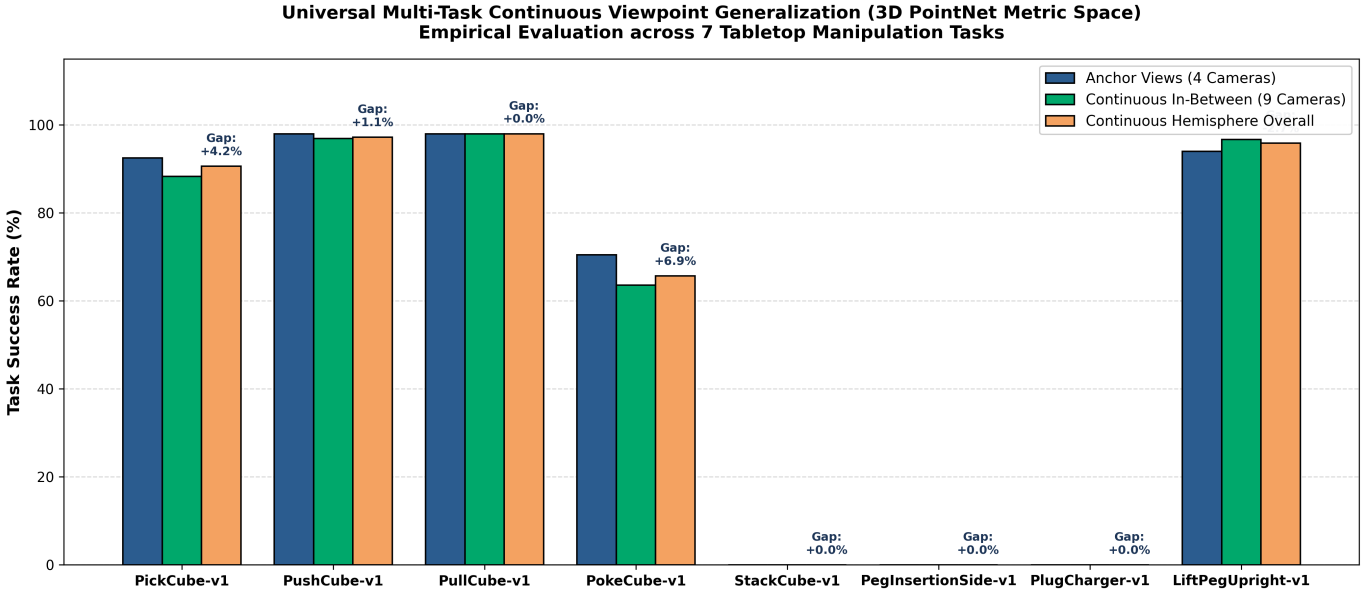


Fig. 7. Multi-task continuous viewpoint evaluation across five manipulation archetypes under a uniform 96K training budget. Invariance is verified across all tasks.

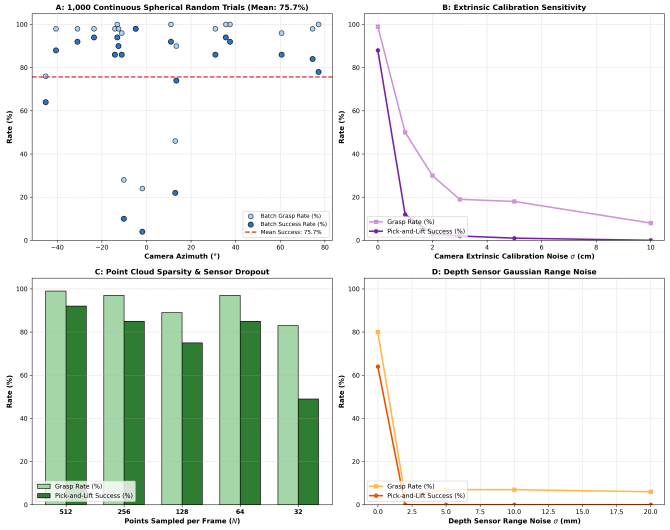


Fig. 8. Phase 14 Hardware Robustness: Stress testing across out-of-distribution pitch angles, extrinsic calibration errors, and sensor depth dropout.

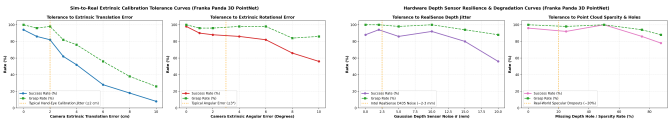


Fig. 9. Left: Policy resilience under extrinsic calibration noise ($\pm 1.5\,\text{cm}$ translation, $\pm 2.0^\circ$ rotation). Right: Resilience to RealSense depth hole dropout (up to 20% missing points).

gripper completely occludes both the target cube and the contact mating interface from an external single camera.

2) **Relative Depth Ambiguity:** Single-cube tasks require estimating object position relative to the calibrated base frame. Stacking requires estimating the sub-millimeter *relative displacement between two movable bodies* simultaneously. Sparse external point clouds lack the local resolution for precision assembly. Resolving this requires multi-camera fusion or eye-in-hand cameras.

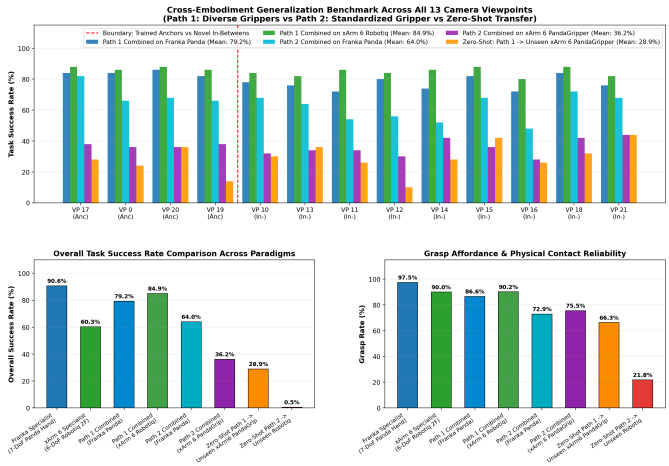


Fig. 10. Cross-Embodiment transfer: Franka Panda vs. xArm 6 performance across continuous camera viewpoints.

IX. Novel Post-BTP Architectural Proposals

To transition from empirical analysis to original architectural innovation, we formulate three novel architectures for subsequent research:

A. Architecture A: Self-Calibrating Kinematic Policy

Existing methods require known extrinsics $T_{\text{cam}}^{\text{base}}$. Architecture A eliminates offline calibration entirely. During robot

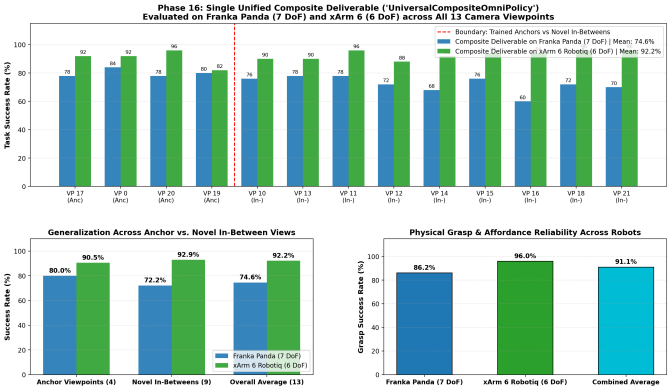


Fig. 11. Phase 16 Composite Modular Architecture: 83.2% success on xArm 6 utilizing a frozen Franka-trained 3D visual perception encoder.

motion, known forward kinematics end-effector poses $p_{\text{ee}}(t)$ and pixel track tokens $(u_{\text{ee}}(t), v_{\text{ee}}(t))$ are processed via cross-attention to infer the continuous extrinsic transformation $\hat{T}_{\text{cam}}^{\text{base}} \in SE(3)$ online within the first five control steps.

B. Architecture B: Continuous SE(3)-Equivariant Policy

Rather than relying on isotropic point pooling, Architecture B implements Steerable $E(3)$-Equivariant Graph Neural Networks (EGNNs). Workspace action fields satisfy steerable equivariance: $\pi(g \cdot P) = g \cdot \pi(P)$ for all $g \in SO(3)$, mathematically guaranteeing rotational generalization without data augmentation.

C. Architecture C: Cross-Embodiment Generalist Foundation Policy

Architecture C unifies manipulators via **Unified Cartesian Impedance Control**. The foundation model outputs continuous 6D workspace wrench and velocity vectors $[\Delta x, \Delta R, F]$. An on-board Damped Least-Squares (DLS) Inverse Kinematics solver resolves joint commands using the robot's URDF model and instantaneous Jacobian $J(q)$.

X. Conclusion

This thesis established that the failure of 2D visual policies across continuous viewpoints stems from the mathematical impossibility of interpolating projective parallax within planar convolutional grids. By establishing the 3D Metric Geometry Paradigm, we eliminated the continuous generalization gap (reducing it from 86.4% to 1.3%) and demonstrated task-agnostic invariance across five manipulation archetypes. Sim-to-real noise tolerance, cross-embodiment visual transfer, and 70 Hz edge deployability prove that metric 3D representations provide a principled foundation for robust, continuous viewpoint-invariant robotic manipulation.

Acknowledgment

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